

# KILLARA HIGH SCHOOL



## YEAR 12 EXTENSION 1 MATHEMATICS

### TERM 2 ASSESSMENT 2010

**Time allowed: 65 Minutes**

#### **Instructions**

1. Show all necessary working.
2. Marks may be deducted for a lack of working or poor setting out of solutions.
3. Start each question in a new booklet.

**Question 1** (15 marks) (Start a new booklet)

- a) Find the value of each of the following. Give your answers in radians in exact form:

i.  $\sin(\cos^{-1}\frac{4}{5})$  1

ii.  $\tan^{-1}\frac{5}{12} + \cos^{-1}\frac{5}{13}$  1

- b) i. State the domain and range of  $y = 3 \sin^{-1}2x$ . 2

- ii. Sketch the function  $y = 3 \sin^{-1}2x$ . 2

- c) Differentiate  $2\sin^{-1}(e^{3x})$  with respect to  $x$ . 2

- d) Consider the function  $f(x) = \frac{4}{x^2 + 4}$ .

Calculate the exact area bounded by the curve and the

$x$ -axis from  $x = -2\sqrt{3}$  to  $x = \frac{2}{3}\sqrt{3}$ . 3

- e) If  $y = \tan^{-1}x$ , then find the value of  $(1+x^2)\frac{d^2y}{dx^2} + 2x\frac{dy}{dx}$  2

- f) Evaluate  $\int \frac{dx}{x^2 + 4x + 5}$ . 2

**Question 2** (16 marks) (Start a new booklet)

a) Differentiate each of the following with respect to  $x$ .

i.  $e^{-2x}\sqrt{x^3}$ . 2

ii.  $\sin[\ln(2x - 3)]$ . 2

b) Integrate  $\int \frac{3\sin 2x}{4 + 5\cos 2x} dx$  with respect to  $x$ . 2

c) Consider the function  $f(x) = \frac{e^x}{3 + e^x}$

i. Show that  $f(x)$  has no stationary points. 2

ii. Find the coordinates of the point of inflexion.

given that  $f''(x) = \frac{3e^x(3 - e^x)}{(3 + e^x)^3}$ . 3

iii. Show that  $0 < f(x) < 1$  for all  $x$ . 3

iv. Sketch the curve of  $y = f(x)$ . Show its essential features. 2

**Question 3** (20 marks) (Start a new booklet)

- a) A hollow cone with a vertical angle of  $60^\circ$  is held with its axis vertical and its vertex downwards. Water is poured into the cone at a rate of  $12 \text{ cm}^3/\text{s}$ . Find the rate at which the surface of the water is rising when the depth is 9cm.  
Give your answer as an exact value. 3
- b) A particle is moving according to  $x = 4\cos\frac{\pi}{4}t$ , in units of metres per second.
- i. Find  $\dot{x}$  and  $\ddot{x}$  2
- ii. Starting at  $x = 0$  sketch the graph of  $x$  showing one complete period. 2
- iii. What are the particles maximum displacement, velocity and acceleration and when during the first 8 seconds do they occur? 3
- iv. How far does it travel during the first 20 seconds? 2

**Question 3 continues over the page**

**Question 3** continued

- c) A particle moving along the  $x$ -axis starts at the origin with an initial velocity  $u$ . Its acceleration is given

by  $\frac{d^2x}{dt^2} = 4x^3 - 16x$ .

- i. Show that the quantity  $E = \frac{1}{2}\left(\frac{dx}{dt}\right)^2 - x^4 + 8x^2$  does not change with time. 3
- ii. Find  $E$  if  $u = \sqrt{\frac{31}{8}}$ . 1
- iii. Show that the particle remains at all times in the interval  $|x| \leq \frac{1}{2}$ . 4
-

## STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

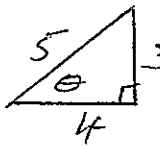
$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

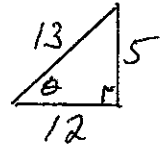
$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left( x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left( x + \sqrt{x^2 + a^2} \right)$$

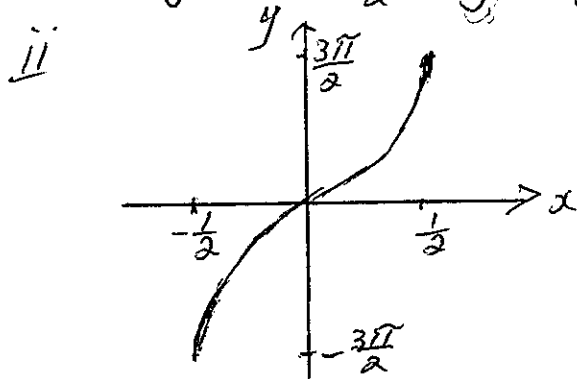
NOTE :  $\ln x = \log_e x, \quad x > 0$

# Question 1

a) i.   $\theta = \cos^{-1} \frac{4}{5}$   
 $\sin(\cos^{-1} \frac{4}{5}) = \sin \theta = \frac{3}{5} \checkmark$

ii.   $\tan \theta = \frac{5}{12}$ ,  $\cos^{-1}(\frac{12}{13}) = 90^\circ - \theta$   
 $\tan^{-1}(\frac{5}{12}) + \cos^{-1}(\frac{12}{13})$   
 $= \theta + 90^\circ - \theta$   
 $= 90^\circ \checkmark$

b) i. Domain:  $-1 \leq 2x \leq 1$   
 $y = 3 \sin x$   $-\frac{1}{2} \leq x \leq \frac{1}{2} \checkmark$   
 Range:  $-\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2} \checkmark$



1 for shape  
 1 for showing

x, y-axis  
 values

c.  $\frac{d}{dx} 2 \sin^{-1}(e^{3x})$   
 $= \frac{2}{\sqrt{1-(e^{3x})^2}} \times 3e^{3x} \checkmark$   
 $= \frac{6e^{3x}}{\sqrt{1-e^{6x}}} \checkmark$



Area =  $\int_{-2\sqrt{3}}^{\frac{2}{3}\sqrt{3}} \left( \frac{4}{x^2+4} \right) dx \checkmark$

d) Continued  $\frac{2}{3}\sqrt{3}$   
 Area =  $2 \int_{-2\sqrt{3}}^{\frac{2}{3}\sqrt{3}} \frac{2}{2^2+x^2} dx$   
 $= \left[ 2 \tan^{-1} \frac{x}{2} \right]_{-2\sqrt{3}}^{\frac{2}{3}\sqrt{3}} \checkmark$   
 $= 2 \tan^{-1} \frac{\sqrt{3}}{3} - 2 \tan^{-1}(-\sqrt{3})$   
 $= 2 \tan^{-1} \frac{\sqrt{3}}{3} + 2 \tan^{-1} \sqrt{3}$   
 $= 2\pi \checkmark$

e)  $y = \tan^{-1} x$   
 $\frac{dy}{dx} = \frac{1}{1+x^2}$   $\frac{d^2y}{dx^2} = -1(1+x^2)^{-2} \cdot 2x$   
 $= (1+x^2)^{-1}$   $= \frac{-2x}{(1+x^2)^2}$

$(1+x^2) \frac{d^2y}{dx^2} + 2x \frac{dy}{dx}$   
 $= (1+x^2) \left( \frac{-2x}{(1+x^2)^2} \right) + 2x \left( \frac{1}{1+x^2} \right) \checkmark$   
 $= \frac{-2x}{1+x^2} + \frac{2x}{1+x^2}$   
 $= 0 \checkmark$

f.  $\int \frac{dx}{x^2+4x+5}$   
 $= \int \frac{dx}{x^2+4x+4+1}$   
 $= \int \frac{dx}{(x+2)^2+1} \checkmark$   
 $= \tan^{-1}(x+2) + C \checkmark$

# Question 2

a i  $\frac{d}{dx} e^{-2x} \sqrt{x^3}$   
 $= \frac{d}{dx} (e^{-2x} x^{\frac{3}{2}})$   
 $= \frac{3}{2} e^{-2x} x^{\frac{1}{2}} + x^{\frac{3}{2}} (-2) e^{-2x}$   
 $= x^{\frac{1}{2}} e^{-2x} \left[ \frac{3}{2} - 2x \right]$

ii  $\frac{d}{dx} \sin[\ln(2x-3)]$   
 $= \cos[\ln(2x-3)] \cdot \left( \frac{2}{2x-3} \right)$

b  $\int \frac{3 \sin 2x}{4 + 5 \cos 2x} dx$   
 $= \frac{3}{2 \times 5} \int \frac{-2 \sin 2x}{\frac{4}{5} + \cos 2x} dx$   
 $= -\frac{3}{10} \ln(4 + 5 \cos 2x) + c$

c i  $f(x) = \frac{e^x}{3 + e^x}$   
 $f'(x) = \frac{3e^x}{(3 + e^x)^2}$   
 $3e^x \neq 0 \therefore f'(x) \neq 0$   
 $\therefore$  no stationary pts

ii  $f''(x) = 0$   
 when  $3e^x(3 - e^x) = 0$   
 ie when  $e^x = 3$   
 ie at  $(\ln 3, \frac{1}{2})$   
 at  $(\ln 3, \frac{1}{2})$

|        |      |      |       |
|--------|------|------|-------|
| x      | 1    | ln 3 | 1.2   |
| f''(x) | 0.01 | 0    | -0.01 |

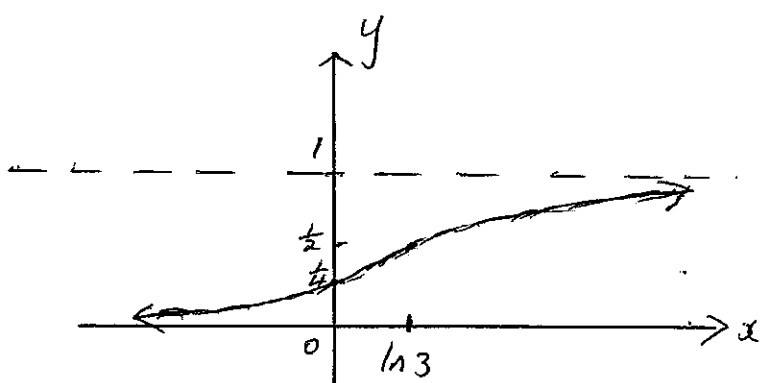
$f''(x)$  changes sign at  $(\ln 3, \frac{1}{2})$   
 and  $f''(x) = 0$  at  $(\ln 3, \frac{1}{2})$   
 $\therefore (\ln 3, \frac{1}{2})$  is the point of inflexion.

iii  $f(x) = \frac{e^x}{3 + e^x}$

$e^x > 0$   
 and  $3 + e^x > e^x$   
 $\therefore \frac{e^x}{3 + e^x} < 1$   
 $\therefore 0 < f(x) < 1$

iv when  $x = 0, f(x) = \frac{1}{4}$   
 $f(x) = \frac{1}{3e^{-x} + 1}$

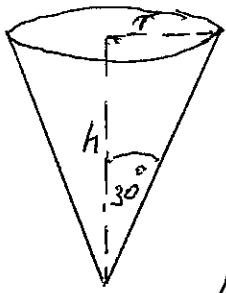
As  $x \rightarrow \infty, e^{-x} \rightarrow 0, f(x) \rightarrow 1$   
 As  $x \rightarrow -\infty, e^{+x} \rightarrow 0, f(x) \rightarrow 0$



1 for shape  
 1 for values (features)

### Question 3

9



$$V = \frac{1}{3} \pi r^2 h$$

$$\frac{dV}{dt} = 12 \text{ cm}^3/\text{s}$$

$$\tan 30 = \frac{r}{h}$$

$$\frac{r}{h} = \frac{1}{\sqrt{3}}$$

$$r = \frac{h}{\sqrt{3}}$$

$$V = \frac{1}{3} \pi \left( \frac{h}{\sqrt{3}} \right)^2 h$$

$$V = \frac{\pi}{9} h^3$$

$$\frac{dV}{dh} = \frac{\pi}{3} h^2$$

$$\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$$

when  $\frac{dV}{dh} = \frac{\pi}{3} h^2$ ,  $\frac{dV}{dt} = 12$ ,  $h = 9$

then

$$12 = 27\pi \times \frac{dh}{dt}$$

$$\frac{dh}{dt} = \frac{12}{27\pi} = \frac{4}{9\pi} \text{ cm/s}$$

10

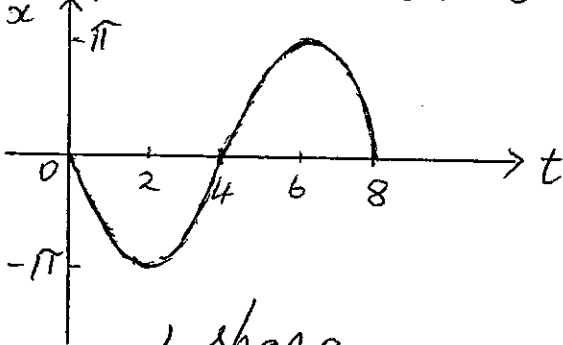
$$x = 4 \cos \frac{\pi}{4} t$$

$$\dot{x} = -\pi \sin \frac{\pi}{4} t$$

$$\ddot{x} = -\frac{\pi^2}{4} \cos \frac{\pi}{4} t$$

$$\dot{x} = -\pi \sin \frac{\pi}{4} t$$

Amp =  $\pi$ , Period = 8



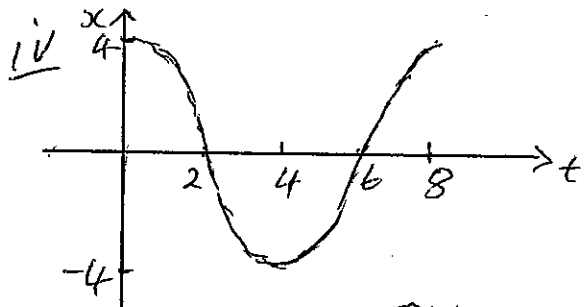
1 shape  
1 values

11

$$\text{Max Displace} = 4 \text{ m when } t = 0 \text{ s and } t = 8 \text{ s}$$

$$\text{Max Vel} = \pi \text{ m/s when } t = 6 \text{ s}$$

$$\text{Max accel} = \frac{\pi^2}{4} \text{ m/s}^2 \text{ when } t = 4 \text{ s}$$



$$x = 4 \cos \frac{\pi}{4} t$$

from graph it travels 8m in 4 seconds

Thus travels  $5 \times 8 = 40$  metres in 20 seconds

12

$$\frac{d^2x}{dt^2} = 4x^3 - 16x$$

$$\frac{d}{dx} \left( \frac{v^2}{2} \right) = 4x^3 - 16x$$

$$\int \frac{d}{dx} \left( \frac{v^2}{2} \right) dx = \int (4x^3 - 16x) dx$$

$$\frac{v^2}{2} = x^4 - 8x^2 + C$$

$$\frac{1}{2} \left( \frac{dx}{dt} \right)^2 = x^4 - 8x^2 + C$$

$$\frac{1}{2} \left( \frac{dx}{dt} \right)^2 - x^4 + 8x^2 = C = E$$

Thus E is a constant which does not change with time.

### Question 3 (continued)

c

ii when  $t=0$ ,  $x=0$   
and  $v=u$

$$E = \frac{1}{2}v^2 - x^4 + 8x^2$$

$$= \frac{1}{2}u^2$$

$$= \frac{1}{2}\left(\sqrt{\frac{31}{8}}\right)^2$$

$$= \frac{31}{16} \quad \checkmark$$

$$\text{iii} \quad \frac{1}{2}v^2 - x^4 + 8x^2 = \frac{31}{16}$$

$$\frac{v^2}{2} = x^4 - 8x^2 + \frac{31}{16}$$

$$\text{since } v^2 \geq 0$$

$$\text{then } x^4 - 8x^2 + \frac{31}{16} \geq 0 \quad \checkmark$$

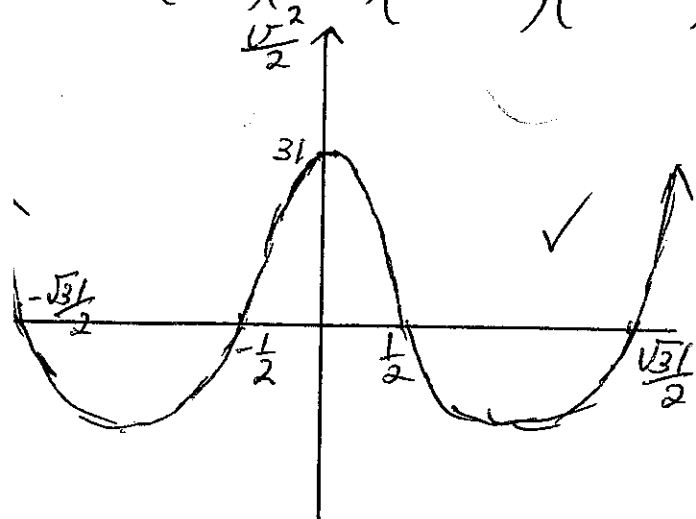
$$16x^4 - 128x^2 + 31 \geq 0$$

$$(4x^2 - 1)(4x^2 - 31) \geq 0$$

$$(2x-1)(2x+1)(2x-\sqrt{31})(2x+\sqrt{31}) \geq 0 \quad \checkmark$$

graph of

$$\frac{v^2}{2} = (2x-1)(2x+1)(2x-\sqrt{31})(2x+\sqrt{31})$$



$$\text{However } \frac{v^2}{2} \geq 0$$

this occurs when  $|x| \leq \frac{1}{2}$

$$\text{and } |x| \geq \frac{\sqrt{31}}{2}$$

However, the particle starts at  $x=0$  when  $t=0$  thus the particle is in the interval  $|x| \leq \frac{1}{2}$   $\checkmark$

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